Growth and Differentiation: Building an Organism

LAUNCH · Science · 40 minutes

I can describe how animals and plants grow and explain how differentiation produces cells specialised for particular functions.

Prepare before pupils arrive

Setup: Prepare fictional/anonymised growth data, process cards and accessible root/tissue images. Avoid personal growth comparisons and pre-teach that most specialised cells share a genome.

Safety: This is a model/data lesson with no tissue culture. Keep health and development discussion non-diagnostic, protect pupil privacy and redirect personal medical concerns to the school's approved support route.

Equipment: Process cards, fictional/anonymised growth table, calculator, comparison frame and optional root-zone or tissue images.

Safety: No biological material is required. Use only fictional or anonymised growth records; do not ask pupils to disclose, compare or diagnose personal height, mass, puberty or health information.

Fallback: Use tactile process symbols, enlarged tables, calculator support and exact pupil-directed scribing. Preserve the same comparison, calculation and explanation.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Edexcel GCSE Biology 1BI0 Foundation · Paper 1 · Topic 2 · specification points 2.5–2.7, growth and cell differentiation in animals and plants.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 4 min	State one role of mitosis in a multicellular organism. Distinguish an increase in cell number from an increase in cell size.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today’s answer.
Starter · 4 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask ‘What makes you think that?’ and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 5 min	growth — increase in size or cell number elongation — cells become longer differentiation — cells become specialised Build an animal-growth chain: mitosis increases cell number; daughter cells grow; some differentiate into specialised tissue cells. Build a plant-growth chain: mitosis in meristem regions increases cell number; cells can elongate and then differentiate.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 5 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 4 min	Read a fictional growth record by naming the variables, units and time interval before describing the trend. Calculate change as final value minus initial value, and percentage change as change divided by initial value multiplied by 100.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each observation to cell division, cell elongation or differentiation, then build one complete animal or plant explanation. Read one tissue observation.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 10 min	Create and apply a comparison model of animal and plant growth, including one growth-data calculation and one evidence-bounded explanation. Record: A two-column animal/plant growth model, one shown calculation, a data description and an explanation that distinguishes direct evidence from cellular inference.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Compare animal and plant growth using one similarity and one difference. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Give one next move; the pupil edits or re-attempts on the existing work.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Both use cell division and differentiation; cell elongation is a particularly important part of plant growth.

Correct. Both organisms make new cells by mitosis and produce specialised cells; elongation also contributes strongly to plant growth.

Misconception repair

Repaired. Differentiation changes gene activity and cell structure while most body cells retain the same genetic information.

Guided checkpoint

Correct. Both organisms make new cells by mitosis and produce specialised cells; elongation also contributes strongly to plant growth. Recall the root sequence: divide near the tip, elongate, then differentiate.

More cells near root tip

CELL DIVISION. Division: mitosis increases the number of cells available for root growth. Ask whether the observation changes cell number, cell length or cell function.

Skin repair cells

CELL DIVISION. Division: increased cell number supports tissue repair. The evidence is a change in the number of cells.

Longer root-zone cells

CELL ELONGATION. Elongation: individual plant cells become longer and contribute to root extension. Use the measured variable: mean cell length, not cell number.

Shoot cells extend

CELL ELONGATION. Elongation: increased cell length contributes to shoot extension. The cells are observed becoming longer rather than producing two daughters.

Xylem structures form

DIFFERENTIATION. Differentiation: cell structure changes to support water transport. This evidence concerns specialised structure and function.

Neurone structures form

DIFFERENTIATION. Differentiation: specialised structure supports rapid communication. Ask what new function the changed cell structure supports.

Model limitation

Block and arrow models separate processes that overlap in living tissues. They omit signalling, nutrition, hormones and cell death, and a growth curve cannot show individual cells dividing or differentiating.

Independent evidence

A two-column animal/plant growth model, one shown calculation, a data description and an explanation that distinguishes direct evidence from cellular inference.

Supported: Complete an animal-versus-plant process table, calculate one simple change and use a because sentence to explain one observation. Choose from division, elongation and differentiation; use final − initial with a pre-set calculator sequence.

Standard: Compare animal and plant growth, calculate percentage change from a supplied dataset and explain one trend using cell processes. Use COMPARE = similarity plus difference; show equation, substitution and unit for the calculation.

Stretch: Evaluate how far organism-level growth data support claims about division, elongation and differentiation, then propose one additional measurement. Separate observation from inference and name the model/data limitation before extending the conclusion.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_L_W10L1_Growth_And_Differentiation_Introduce.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.